HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Repetition to

ADVANCED TOPICS IN ENGINEERING PHYSICS

‘Plasma and its applications’

Prof. Dr. Christoph Gerhard, 2024

1 Basics of plasma physics

- 1. Give an approximate value for the electron density of the ionosphere.
- 2. The colour of polar lights is usually either greenish or purple. Why?
- 3. Describe – in short – the 0th and the 5th state of aggregation.
- 4. Are all plasmas fully ionised?
- 5. What are positive and negative charge carriers within plasmas?
- 6. What is the so-called ‘Crookes dark space’?
- 7. What are cathode rays?
- 8. What are canal rays?
- 9. Which interactions are dominant within plasmas, interactions between two charged particles or interactions between a charged particle and a neutral one…and why?

- 10. How many ions and electrons can be found within a gas at ambient conditions?
- 11. Thermal ionisation of atoms and dissociation of molecules starts at a temperature of…?
- 12. What is a ‘non-equilibrium plasma’?
- 13. Name at least three processes or effects that occur within plasmas.
- 14. What is – roughly – the electron density of a so-called glow discharge?
- 15. Name at least two examples for a) plasma-chemical and b) plasma-physical mechanisms within plasmas.
- 16. Explain shortly the function of a fluorescent lamp.
- 17. Do the electrodes of a glow discharge plasma source emit any electrons?

- 18. Above which temperatures does thermoemission, aka thermionic emission, occur?
- 19. Name at least three parameters that define the type or kind of a plasma discharge.
- 20. Name at least three characteristic properties of plasmas.
- 21. Name at least three essential plasma parameters.
- 22. Give a definition of ‘quasi-neutrality’.
- 23. What is the difference between quasi-neutrality and strict neutrality?
- 24. What is the underlying reason or effect for quasi-neutrality?
- 25. What is the so-called ambipolarity of a plasma?
- 26. Shortly explain the effect of Debye shielding.

- 27. Give a definition for the Debye length.
- 28. What is the so-called ‘sheath-effect’?
- 29. In practice, what is a ‘wall’ in context with plasmas?
- 30. What is the difference between ionisation and dissociation?
- 31. Which effect is more probable, ionisation or dissociation?
- 32. What does the Boltzmann-equation express?
- 33. In which cases does the Boltzmann-equation apply?
- 34. What is the mean free length of path?
- 35. Name the pressure range of full vacuum.
- 36. What is the key message of the drift-diffusion-approximation?

- 37. Give a definition for the term ‘drift’.
- 38. The flux or current density of particles can be determined on the basis of which parameters?
- 39. What is diffusion of particles…and what is the underlying reason?
- 40. What is a plasma in thermodynamic equilibrium?
- 41. Which equation is used for the determination of the degree of ionisation?
- 42. Which plasma parameter has the major impact on the degree of ionisation?
- 43. Give a short definition of the plasma frequency.
- 44. What is the difference between a (i) low temperature non-equilibrium plasma and a (ii) low temperature plasma in local thermodynamic equilibrium?
- 45. What is a so-called relativistic plasma?